AKC-ADC(i) vs dki2d_fast

Comparison Report | Series 16 Abdominal DWI | b = 50 / 400 / 800

Data Description

Test volume: series 16 abdominal DWI, shape 268x184x56 voxels, 3 shells (b = 50, 400, 800 s/mm2). No true b0 image available. Models run on full volume with Otsu mask (853,436 valid voxels). All computations single-threaded for fair timing comparison.

Key Finding

D and K maps are bit-identical between both methods (r = 1.000000, MAE = 0.000000).

Parameter Agreement

Metric	D	K	S0
Pearson r	1.000000	1.000000	0.9841
MAE	0.000000	0.000000	15.71
RMSE	0.000000	0.000000	33.73
Max absolute diff	0.000000	0.000000	918.71
Within 10%	100.0%	100.0%	88.4%
Within 20%	100.0%	100.0%	93.6%
Within 30%	100.0%	100.0%	95.8%

Why D and K Are Identical

Both methods solve the same linear system -- the log-linear DKI least squares regression. The only difference is the computational path:

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dki2d_fast:  ln(S) = b0 - b*D + b^2*D^2*K/6  via 3x3 matrix inversion
akc_adci:    ADC(i) = (ln S_i - ln S_{i+1}) / (b_{i+1} - b_i)
             ADC = D + slope * b_mid       via weighted regression
```

The secant ADCs are a linear transformation of the same log-signal data. Both yield the same D and slope because the algebra is equivalent.

S0 Estimation Difference

Method	Mean	Median	Std
dki2d_fast (exp(intercept))	104.22	68.28	110.50
akc_adci (direct min-b read)	88.51	60.00	86.27

dki2d_fast extrapolates to b=0 via the quadratic fit, producing an intercept that is always >= the measured signal at b=50. akc_adci takes the raw signal at b_min. 853,371/853,436 voxels (100.0%) have S0_fast >= S0_akc. Mean relative difference: 11.6%.

Residual Comparison

Method	Mean RMSE	Voxels > 0.02
dki2d_fast	0.0599	28,640 (3.4%)

akc_adci	0.0000	0 (0.0%)

akc_adci reports zero residual because with 3 b-values there are 2 ADC secants, fitting 2 parameters (D, slope) leaves 0 degrees of freedom -- a perfect fit by construction. dki2d_fast reports meaningful residuals from the least-squares fit.

Kurtosis Distribution (identical for both)

Range	Voxels	%
K < 0	0	0.0%
K > 3	119,021	13.9%
K > 5	60,473	7.1%
K >= 10 (clipped)	28,652	3.4%

Timing

Method	Time (s)	Relative
dki2d_fast	149.2	1.0x (baseline)
akc_adci	96.5	1.5x faster

Conclusion

akc_adci produces bit-identical D and K maps to dki2d_fast. The only differences are:

1. S0: direct min-b read vs fitted intercept (systematically ~12% lower)
2. Residual: zero by construction (underdetermined system) vs meaningful RMSE
3. Speed: 1.5x faster (lighter per-voxel computation)

The model works as intended -- a transparent, fast screening tool that produces the same DKI parameters as the existing estimator via a different (algebraically equivalent) computational route.